

Feasibility-Aware Seed Budgeting for Constrained iLQR: Diagnostic Multi-Armed Bandits and Bayesian Optimization

Jie Su

Abstract—Constrained iterative LQR / DDP [1–3] is a workhorse for trajectory optimization in robotics, yet it is acutely sensitive to the initial trajectory: under a tight iteration budget, a poor warm-start yields slow convergence, local minima, or outright infeasibility. A standard remedy is multi-start: draw K seeds, allocate each a uniform budget share, and retain the best feasible trajectory. Uniform allocation is simple—and wasteful—when only a minority of seeds lie in favourable basins. We formulate multi-start constrained iLQR as a sequential budget-allocation problem. After each short warm-started batch, the solver exposes diagnostics—feasible cost improvement ΔJ , constraint violation v , improvement rate ρ , and the closed-loop convergence measure χ [4]—which form the selection signal: the primary reward combines the first three, while χ serves only as a termination switch and as a feature for Bayesian optimisation. We prove that any score in a natural class of local-improvement diagnostics can mis-rank seeds relative to their terminal feasible cost (ranking inversion), so greedy selection by a single score is not universally optimal. We instantiate Uniform, UCB, and GP-EI policies on this foundation. Adaptive allocation with the proposed reward improves over Uniform on a wheeled-navigation and an articulated-arm benchmark (feasible-first comparison on 5/5 random seeds; $\approx 38\%$ lower mean cost), while greedy χ -selection proves unreliable; a contact-rich quadruped template provides a ranking-inversion diagnostic on a third physical domain. The framework is budget-exact; the benchmark definitions and all allocation traces are released, and the scheduling framework and host solver are distributed as a prebuilt binary for independent reproduction.

Index Terms—Trajectory optimization, constrained iLQR, warm-start, multi-armed bandit, Bayesian optimization, model predictive control.

I. INTRODUCTION

MULTI-START trajectory optimization is a common strategy for mitigating the sensitivity of iterative LQR / DDP [1–3] to the initial guess. Given a library of K seed trajectories (drawn, e.g., via LHS [5] or homotopy heuristics [11]), a fixed total budget of B solver iterations, and a uniform allocation rule, each seed receives B/K iterations and the best feasible result is retained [6, 7].

Uniform allocation is simple, but it is wasteful whenever the seeds differ substantially in quality: a seed that starts in a poor basin receives the same budget as one descending rapidly toward a good local optimum. The decision problem is therefore *how to allocate the remaining budget among seeds dynamically, based on observations collected during optimisation*.

The solver exposes several diagnostics after each short batch of b iterations: the feasible cost J , the max constraint violation v , the per-iteration improvement rate ρ , and the closed-loop convergence measure $\chi = \|\prod \hat{A}_k\|_2$ [4], where $\hat{A}_k = A_k - B_k K_k$ are the closed-loop dynamics under LQR

feedback. These signals contain information about the local landscape, but they are not a perfect predictor of the terminal feasible cost J^* that a seed will achieve after full budget. We show formally (Section II-C) that a *ranking inversion* must occur: a score computed from batch diagnostics can prefer a seed whose terminal feasible cost is worse than another that the same score ranks lower. This ranking inversion implies that greedy selection by any fixed score is not universally optimal, and that some form of exploration is needed in principle.

Against this backdrop we formulate multi-start constrained iLQR as a sequential budget-allocation (multi-armed bandit) problem and instantiate three policies—Uniform, UCB [8], and GP-EI (Bayesian optimisation [9, 10])—on a frozen feasibility-improvement reward that combines ΔJ , v , and ρ , with χ reserved as a termination switch and a feature for BO.

Related work. Constrained iLQR/DDP and public solvers [1–3]; direct transcription methods [12, 13]; multi-start and warm-starts [6, 7, 11]; MAB / UCB [8]; BO with GP-EI [9, 10, 14]; the convergence measure χ [4]. Our contribution is a *feasibility-aware budgeting* interface between these literatures and the practical problem of allocating solver iterations across warm-starts.

Contributions.

- 1) **Ranking inversion analysis.** We define the class of local-improvement scores S_{LI} (Definition 1) and prove that any score in this class can disagree with the oracle ordering by terminal feasible cost (Lemma 1). This implies that greedy selection by such scores is not universally optimal (Corollary 1), motivating exploration.
- 2) **Feasibility-aware reward and protocol.** We freeze a scalar primary reward (Equation (3)) that combines feasible cost improvement, constraint violation, and improvement rate, with χ serving only as a termination switch and a BO feature dimension. Uniform, UCB, and GP-EI policies share an exact budget-accounting scheduler with warm-start-only batches.
- 3) **Empirical findings on real constrained-iLQR instances.** We demonstrate ranking inversion (Corollary 1) on wheeled-navigation and contact-rich quadruped benchmarks: both the feasible-improvement score and $-\log \chi$ mis-rank seeds against the oracle ordering (Tables IV and V). Under the frozen protocol, adaptive allocation (UCB, BO-EI, Greedy- s_r) wins the feasible-first comparison against Uniform on all five RNG seeds of the navigation benchmark (one-sided sign test, $p = 2^{-5}$) and attains $\approx 38\%$ lower mean cost on an all-feasible arm benchmark, while greedy χ -selection is unreliable in both settings.

Scope. The paper does not claim solver superiority: the host is a constrained iLQR stack validated for parity with open-source solvers on a standard gate suite (Section III-B), and the scheduling claims concern iteration-budget allocation. The quadruped template serves as a ranking-inversion diagnostic, not as a policy-separation suite. Adaptive policies (UCB, BO-EI, Greedy- s_r) may tie with each other; we do not claim to separate them on the current seed libraries. The implementation is C++17. The benchmark definitions, experimental protocol, and all allocation traces are released; the seed-budgeting scheduler and the constrained-iLQR host are distributed as a prebuilt binary (Docker image) to enable independent reproduction.¹

II. METHOD: FEASIBILITY-AWARE SEED BUDGETING

A. Background: Constrained iLQR and Diagnostics

Consider a discrete-time optimal control problem with horizon N , state dimension n , and control dimension m :

$$\begin{aligned} \min_{\{x_k, u_k\}} \quad & \ell_N(x_N) + \sum_{k=0}^{N-1} \ell_k(x_k, u_k), \\ \text{s.t.} \quad & x_{k+1} = f_k(x_k, u_k), \\ & g_k(x_k, u_k) \leq 0, \quad \forall k \in [0, N-1], \\ & h_k(x_k, u_k) = 0, \quad \forall k \in [0, N-1], \\ & g_N(x_N) \leq 0, h_N(x_N) = 0. \end{aligned} \quad (1)$$

where $\ell_k(x_k, u_k)$ and $\ell_N(x_N)$ are the stage and terminal cost functions, $f_k(x_k, u_k)$ is the nonlinear dynamics, $g_k(x_k, u_k)$ and $h_k(x_k, u_k)$ are inequality constraints (control limits, joint limits, etc.), and $h_N(x_N)$ are equality constraints (waypoints, terminal pose, etc.). Constrained iLQR solves (1) via iterative linearization. At each iteration, the dynamics and cost are quadratized around the current nominal trajectory $\{\bar{x}_k, \bar{u}_k\}_{k=0}^N$, yielding a local LQR subproblem whose solution is obtained via backward Riccati recursion, producing feedback gains K_k and feedforward terms d_k . Constraints are handled with an ADMM / Filter line-search stack and optional Box-QP control limits in the Riccati step (implementation details in the host solver; see the reproducibility footnote). The forward pass applies $\delta u_k = K_k \delta x_k + \alpha d_k$ with filter or backtracking line search.

The backward pass also yields the closed-loop convergence measure [4]:

$$\chi = \|\Phi\|_2, \quad \Phi = \prod_{k=N-1}^0 \tilde{A}_k, \quad \tilde{A}_k = A_k - B_k K_k \quad (2)$$

where $A_k = \partial f_k / \partial x$, $B_k = \partial f_k / \partial u$, and K_k are the LQR gains. The matrix $\tilde{A}_k$ is the closed-loop state transition matrix, and Φ propagates perturbations over the full horizon. $\chi < 1$ indicates exponential decay of trajectory perturbations (local basin of attraction), while $\chi \gg 1$ signals poor convergence. Computation requires one $O(Nn^3)$ matrix chain multiplication—performed automatically at each backward pass with negligible overhead. We return to the properties of χ in Section II-C2.

B. Seeds as Bandit Arms

We cast multi-start iLQR as a multi-armed bandit [8]: arm i is a seed trajectory $\tau^{(i)}$. At round t we select i_t , run a batch of b iLQR iterations with *trajectory* warm-start (a fresh solver instance; dual variables are not assumed to persist), and observe diagnostics $(J_{\text{before}}, J_{\text{after}}, v, \chi)$, where v is the max constraint violation over all stages.

1) *Primary reward design:* The reward encodes the decision objective: under a tight budget, prefer seeds that are feasible and whose feasible cost is still decreasing, while avoiding credit for cost reduction achieved under infeasibility. Define

$$\begin{aligned} \Delta J &= J_{\text{after}} - J_{\text{before}}, \\ \text{viol}_+ &= \max(0, v - \varepsilon), \\ \Delta J_{\text{feas}} &= \begin{cases} \Delta J, & v \leq \varepsilon, \\ 0, & \text{otherwise,} \end{cases} \\ \rho &= \frac{\max(0, -\Delta J_{\text{feas}})}{\max(1, b_{\text{used}})}. \end{aligned}$$

The primary reward is

$$r_t^{(i)} = \text{clip}\left(-\Delta J_{\text{feas}} - \lambda \text{viol}_+ + \eta \rho\right), \quad (3)$$

with weights $\lambda, \eta > 0$. The three terms serve distinct roles:

- $-\Delta J_{\text{feas}}$ is the main signal, active only when the arm ends the batch feasible. It rewards feasible cost reduction; an infeasible arm receives zero from this term.
- $-\lambda \text{viol}_+$ penalises constraint violations linearly, remaining active whether or not the arm is feasible.
- $\eta \rho$ is a per-iteration rate bonus. When the solver consumes fewer iterations than the batch size b (early convergence or numerical failure), this term amplifies the improvement rate slightly.

2) *Role of χ :* The convergence measure χ is *not* inserted into (3). Instead it serves two auxiliary purposes:

- **Termination switch.** If $\chi > \chi_{\text{cut}}$ or the batch fails numerically, arm i is suspended and removed from further allocation, preventing budget from being wasted on seeds whose local linearisation is unstable.
- **BO feature.** When Bayesian optimisation is used as the scheduling policy, $\log_{10} \chi$ is included as a dimension of the seed feature vector (Section II-E), giving the GP surrogate access to convergence information without making χ the objective.

We also optionally stop the entire allocation once a feasible incumbent reaches a declared near-optimal cost threshold (early success), avoiding expenditure of residual budget on poor seeds.

C. Theoretical Motivation: Why Instantaneous Arm Scores Are Not Fixed

The scheduler observes a diagnostic score $s(d_i(t))$ after each warm-started batch and must decide which seed receives further iterations. Exploration (UCB / BO) is justified if greedy

¹<https://github.com/jiesu-zju/optimal-solver>

ranking by s can disagree with the oracle ranking by terminal feasible cost J_i^* . We establish this disagreement through a core lemma on local-improvement scores (Section II-C1), an auxiliary analysis of χ (Section II-C2), and a decomposition of the primary reward s_r that delimits its regime of applicability (Section II-C3).

1) *Local-improvement scores can mis-rank seeds*: We first define the class of scores to which our main result applies.

Definition 1 (Local-improvement score class). *A score s belongs to S_{LI} if it depends only on the diagnostics of a single warm-started batch and, for any two pulls where both arms finish feasible ($v \leq \varepsilon$), s is strictly increasing in the negative cost change $-\Delta J$ (larger feasible decrease $\Rightarrow$ higher score).*

The dominant term of our primary reward s_r —namely $-\Delta J_{\text{feas}}$ when $v \leq \varepsilon$ —belongs to S_{LI} .

Lemma 1 (Local-improvement incompleteness). *For any $s \in S_{LI}$, there exist instances with two arms sharing the same initial feasible cost J_0 such that*

$$s(d_A(1)) > s(d_B(1)) \quad \text{but} \quad J_A^* > J_B^*,$$

where J_i^* is the terminal feasible cost of arm i under equal remaining budget after round 1.

Proof. Fix $s \in S_{LI}$, common initial cost J_0 , and choose any $a > b > 0$ as the first-batch cost decreases of arms A and B , respectively. Both batches finish feasible by construction, so $s(d_A(1)) > s(d_B(1))$ follows from $s \in S_{LI}$.

Specify the second-batch decreases $c, d \geq 0$ as

$$\begin{aligned} J_A &: J_0 \xrightarrow{-a} J_0 - a \xrightarrow{-c} J_0 - a - c, \\ J_B &: J_0 \xrightarrow{-b} J_0 - b \xrightarrow{-d} J_0 - b - d. \end{aligned}$$

The terminal costs are $J_A^* = J_0 - a - c$ and $J_B^* = J_0 - b - d$. The inequality $J_A^* > J_B^*$ is equivalent to $a + c < b + d$, i.e. $d - c > a - b > 0$.

Since $a - b$ is a fixed positive number, choose $c = 0$ (arm A makes no further progress after the first batch—a shallow basin) and any $d > a - b$ (arm B continues to improve—a deep basin). These choices are always feasible, giving $s(d_A(1)) > s(d_B(1))$ with $J_A^* > J_B^*$. ■ □

Corollary 1 (Greedy insufficiency of single-batch local-improvement scores). *No policy that greedily selects the arm with the highest current score $s \in S_{LI}$ can be universally optimal with respect to terminal feasible cost J^* .*

Remark 1 (Scope). *Corollary 1 does not rule out the possibility that a richer score aggregating multi-batch history (e.g. a running average or a GP surrogate) could recover the oracle ordering. It shows only that a single batch of local-improvement information is insufficient. This distinction motivates the construction of multi-batch features in the BO policy (Section II-E) and the exploration bonus in UCB—both of which accumulate information across pulls.*

Remark 2 (Specialisation to s_r). *The dominant term of our primary reward s_r (Equation (3)) is $-\Delta J_{\text{feas}}$. When the batch finishes feasible ($v \leq \varepsilon$), this term belongs to S_{LI} and therefore inherits the ranking inversion of Lemma 1. The second term*

$-\lambda \text{viol}_+$ is active only when the arm is infeasible; it is not covered by this lemma and is examined empirically in Section II-C3.

Remark 3 (Generality). *Lemma 1 constructs an abstract pair of cost sequences; it does not guarantee that every constrained iLQR problem realises such a configuration. Whether a concrete problem and seed library produce the required “fast-into-shallow-basin” and “slow-into-deep-basin” pair is an empirical question. Scores outside S_{LI} (e.g. pure violation penalties) could in principle avoid the construction altogether.*

The lemma suffices because $-\Delta J_{\text{feas}} \in S_{LI}$ is the dominant term driving our protocol’s seed selection; the empirical evidence in Section III-C confirms that the required conditions arise in practice.

2) *Note on χ* : Recall $\chi = \|\prod \tilde{A}_k\|_2$ from Section II-A. The spectral norm of a matrix product is not monotone in the norms of the individual factors (e.g. $\text{diag}(10, 0.01) \cdot \text{diag}(0.01, 10) = \text{diag}(0.1, 0.1)$ has norm 0.1, while tightening each factor to $\text{diag}(9.9, 0.01)$ yields a product with norm 98.01). Therefore χ is not a quantity that behaves monotonically with local cost improvement—it can increase when individual linearizations appear to contract. In our protocol χ serves only as a termination switch (arms with diverging χ are suspended) and as a BO feature dimension.

3) *Reward decomposition and regime of applicability*: The primary reward $s_r = -\Delta J_{\text{feas}} - \lambda \text{viol}_+ + \eta\rho$ has three components whose roles differ in the ranking-inversion landscape.

Feasible improvement ($-\Delta J_{\text{feas}}$). This is the dominant term driving seed selection when an arm is feasible. It belongs to S_{LI} ; Lemma 1 therefore guarantees that ranking inversion can occur, and greedy selection by this term alone is not universally optimal. Section III-C provides concrete illustrations of this inversion on real constrained-iLQR instances.

Violation penalty ($-\lambda \text{viol}_+$). This term does *not* belong to S_{LI} ; it does not reward cost decrease and is active only in the infeasible regime. It encodes a soft preference: arms that remain persistently infeasible accumulate a negative score that lowers their pull priority. However, the term is not a provable separator between “temporarily infeasible but promising” and “permanently infeasible” arms, because a moderately infeasible arm with superficially good cost can still receive a higher total score than a barely feasible one. Its effectiveness is problem-dependent and examined empirically in the experiments.

Rate bonus ($\eta\rho$). This term provides a small per-iteration improvement bonus that becomes relevant when the solver converges early or fails numerically (so $b_{\text{used}} < b$). Under nominal operation its effect on ranking is minor.

Regime of applicability. Lemma 1 establishes that local-improvement scores—including the dominant term of s_r —are *not* universally reliable predictors of terminal feasible cost. This negative result does not, however, imply that s_r is useless as a bandit reward. Bandit policies (UCB, BO) require only that the reward distribution carries *some* signal about arm quality.

In practice, the regimes where s_r provides a useful signal can be characterised per seed library. After warm-up, if the empirical mean rewards $\bar{s}_r^{(i)}$ (or the feasible incumbent costs) are positively correlated with the oracle ordering J_i^* , then bandit allocation can exploit this signal. When the correlation is weak—for example because warm-up already identifies the best basin, as on the current benchmarks—the adaptive policies remain tied *with each other*, and harder seed libraries are required to separate them; the separation of the adaptive bloc from Uniform on the navigation and arm benchmarks (Section III-D) shows that s_r nevertheless carries exploitable signal in practice. When the correlation is systematically negative (early s_r anti-correlates with J^* and later batches do not correct it), the reward or feature representation must be revised. This per-library verification is part of the experimental protocol rather than a theoretical guarantee; Section III-D provides diagnostic statistics.

D. UCB Policy (Independent Arms)

UCB treats each seed as an independent bandit arm under reward (3). At scheduling round t , the policy selects the seed maximizing:

$$i_t^{\text{UCB}} = \operatorname{argmax}_{i \in \mathcal{A}_t} \left(\hat{r}_{i,t} + c \sqrt{\frac{2 \log t}{n_{i,t}}} \right) \quad (4)$$

where $\mathcal{A}_t \subseteq \{1, \dots, K\}$ is the set of active (non-terminated) seeds, $\hat{r}_{i,t}$ is the empirical mean reward, $n_{i,t}$ is its pull count, and c is an exploration constant. Seeds with $\chi > \chi_{\text{term}}$ are terminated and removed from $\mathcal{A}_t$. The total budget B is the only hyperparameter.

Theoretical justification. With clipped feasible-improvement rewards and χ -based arm suspension, active-arm rewards are bounded. We use $c = 1.0$.

E. BO Policy (GP–EI)

While UCB treats seeds independently, BO models similarity via a Gaussian Process and selects by Expected Improvement [9] on reward (3). We use it as a tool, not a methodological innovation; the description here is kept brief.

Each arm is mapped to a diagnostic feature vector $\phi^{(i)} = [\Delta J, v, \log_{10} \chi, n_i, \bar{\rho}]$, updated after every pull. The arm index is excluded to prevent the GP from learning arbitrary arm identity rather than generalisable features.

We model the reward function over seed feature space as a GP [14] with RBF kernel $k(\phi, \phi') = \sigma_f^2 \exp(-\|\phi - \phi'\|_2^2 / 2\ell^2)$, kernel hyperparameters set to the benchmark values listed in Table II—treated as fixed settings rather than optimised values. The GP posterior mean $\mu(\phi^*)$ and variance $\sigma^2(\phi^*)$ follow the standard closed-form GP equations [14]; exact Cholesky inference costs $O(K^3)$, negligible for $K = 8$.

The next seed is selected by maximising Expected Improvement [9] $\text{EI}(\phi) = (\mu - r_{\text{best}} - \xi) \Phi(Z) + \sigma \varphi(Z)$ with $Z = (\mu - r_{\text{best}} - \xi) / \sigma$ and $\xi = 0.01$, where Φ and φ are the standard normal CDF and PDF.

Algorithm 1 Feasibility-Aware Seed Budgeting

- 1: **Input:** Seeds $\{\tau^{(i)}\}_{i=1}^K$, budget B , batch b , warm-up w , policy $\pi \in \{\text{UCB}, \text{BO}\}$
 - 2: Initialize $n_i \leftarrow 0$, $\hat{r}_i \leftarrow 0$, $\chi_i \leftarrow \infty$ for all i
 - 3: **Phase 1:** Run w warm-started iLQR iterations on each seed; compute diagnostics and r_i via (3); update $n_i \leftarrow n_i + 1$
 - 4: **for** $t = Kw + 1$ **to** B **step** b **do**
 - 5: $i^* \leftarrow \text{SELECT}(\pi)$ // UCB or EI policy
 - 6: Run b warm-started iLQR iterations on seed i^* (using the previous solution as the initial guess); observe χ_{i^*}
 - 7: Observe diagnostics; set r_{i^*} via (3); update $\hat{r}_{i^*}$, n_{i^*}
 - 8: **if** $\chi_{i^*} > \chi_{\text{term}}$ **then**
 - 9: Terminate seed i^*
 - 10: **end if**
 - 11: **end for**
 - 12: **Return** solution of seed with lowest final cost
-

III. EXPERIMENTS

We evaluate feasibility-aware seed budgeting on three frozen instances that collectively cover the paper’s theoretical claims: ranking inversion (Layer 1), policy discrimination under s_r , and the negative control χ -greedy. All experiments use reward (3) with $(\lambda, \eta) = (10, 0.1)$, $\chi_{\text{term}} = 100$, disabled early-success stopping, and the solver parameters of Table II.

A. Setup and Protocol

Table I lists the three benchmarks. Each is run with five random seeds (42–46), and reported metrics are mean $\pm$ std across those seeds unless noted.

Policies. We compare five strategies sharing reward (3): **Uniform** (equal allocation), **UCB** [8] (Eq. (4), $c = 1.0$), **BO–EI** (GP with RBF kernel, Section II-E), **Greedy- s_r** (always pull the seed with the highest current s_r), and **Greedy- χ** (always pull the seed with the lowest χ). The scheduler follows Algorithm 1 with trajectory warm-start and $\chi_{\text{term}} = 100$.

1) *Benchmark descriptions:* **B1: Nav2D–Hard.** Dynamics: bicycle model [4] with state $x = [x, y, \theta, \kappa]^\top$ and control $u = [v, \dot{\kappa}]^\top$, discretised via RK4 ($\Delta t = 0.1$ s, $N = 80$). Cost: waypoint attraction $w_p \|p - p^*\|^2$, curvature penalty $w_\kappa \kappa^2$, control regularisation $w_u \|u\|^2$, plus six circular soft obstacles with quadratic penetration cost $W \max(0, r + m - \|p - c\|)^2$ ($W = 80$, $m = 0.35$ m) forming a dense field with two homotopy classes. Constraints: control limits $|v| \leq 3$, $|\dot{\kappa}| \leq 2$; curvature $|\kappa| \leq 1.5$; terminal position tolerance ± 0.5 m, heading ± 0.3 rad. Setup: start (0, 0), target (10, 0); $K = 8$ seeds from LHS noise ($w = 0.5$, alternating heading bias), $B = 160$, $b = 10$, warm-up 10.

B2: Planar quadruped. Dynamics: planar floating-body template with state $(p_x, p_z, \theta, q_{\text{hip}}^f, q_{\text{knee}}^f, q_{\text{hip}}^b, q_{\text{knee}}^b, \dot{q}) \in \mathbb{R}^{14}$ and control (joint torques) $u \in \mathbb{R}^4$; analytic Jacobians. Cost: quadratic body-pose tracking and joint regularisation; ground interaction and joint limits encoded as soft costs (step penetration penalty, hip retraction barrier, knee-angle bounds) rather than explicit contact forces. Constraints: knee-angle bounds; terminal box on body position $|p_x| \leq 0.2$, $-0.1 \leq p_z \leq 0.4$.

TABLE I
BENCHMARK PROTOCOL.

ID	System	K	Warm-up	Batch	B	Role
B1	Nav2D-Hard (bicycle + dense soft obstacles)	8	10	10	160	Primary policy discrimination
B2	Planar quadruped (soft-contact template) [†]	8	2	5	80	Contact-rich ranking-flip diagnostic
B3	5-DOF planar analytic arm (UP/DOWN homotopy)	8	3	5	80	Second discrimination + clean χ negative control

[†] Diagnostic mode uses $K=4$ seeds with equal per-seed budget (warm-up 2, then +30 iterations) and no bandit; the $K=8$ suite configuration is a near-tie sanity check only.

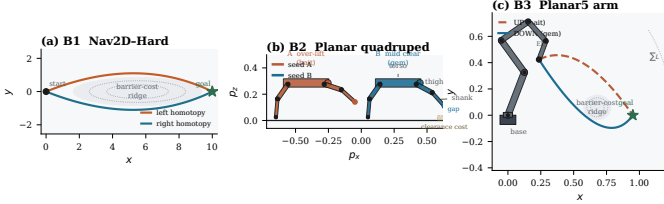


Fig. 1. Benchmark schematics (robot geometry first). Faint dotted mists mark *barrier-cost ridges* in J (formulation detail), not obstacle bodies. (a) Nav2D-Hard: left/right homotopy warm-starts around a mid-field cost ridge. (b) Planar quadruped with torso and thigh/shank capsules; seed A over-lifts the front foot (bait) while seed B keeps a mild clearance above the ground clearance-cost zone (gem). (c) Planar5 solid-link arm in a left-side ready pose; UP (bait) / DOWN (gem) end-effector corridors flank a mid-workspace cost ridge.

TABLE II
SOLVER AND SCHEDULING PARAMETERS.

Parameter	Symbol	Value
Reward: feasible-improvement coefficient	1	(implicit)
Reward: violation weight	λ	10
Reward: rate-bonus weight	η	0.1
UCB exploration constant	c	1.0
χ termination threshold	χ_{term}	100 (never triggered)
Early-success cost threshold	J_{succ}	∞ (disabled)
GP kernel length scale	ℓ	2.0
GP signal variance	σ_f^2	1.0
GP observation noise	σ_n^2	10^{-4}
El hyperparameter	ξ	0.01
ADMM inequality penalty	μ	10
Regularization	λ_{reg}	10^{-4}
Constraint tolerance	ε	10^{-3}
Line search reduction	α_{decay}	0.5
Filter margin	δ	10^{-4}
Filter max size	M	5

Setup: $K = 8$ seeds, $B = 80$, $b = 5$, warm-up 2; for the ranking-flip diagnostic (no bandit), $K = 4$, warm-up 2, +30 additional iterations.

B3: Planar5. Dynamics: 5-DOF planar analytic arm, double-integrator discretisation, state $x \in \mathbb{R}^{10}$, control $u \in \mathbb{R}^5$, exact Jacobians. Cost: quadratic end-effector tracking, joint regularisation, soft disk obstacle. Constraints: joint-limit inequalities; no terminal box. Setup: $K = 8$ seeds split into an *UP* cluster (bait: fast initial descent into a shallow basin) and a *DOWN* cluster (gem: slower initial descent into a deeper terminal basin), directly instantiating Lemma 1; $B = 80$, $b = 5$, warm-up 3.

B. Host Solver

All experiments use the same constrained iLQR stack described in Section II-A. Because the host is distributed as a

TABLE III
GATE VALIDATION OF THE HOST AGAINST THE BEST FEASIBLE OPEN-SOURCE BASELINE (CROCODDY, ALIGATOR, OR ALTRO); FIVE PROBLEMS $\times$ THREE SEEDS, ALL 15/15 INSTANCES FEASIBLE.

Problem	Ours	Best SOTA (ratio)
Pendulum swing-up	186.1	1.000
Double-integrator reach	6.67	1.000
Cartpole stabilise	35–114	≈ 1.000
Quadrotor hover	4431–5477	≤ 1.02
Bicycle navigation	538.9–539.5	≈ 1.000

TABLE IV
RANKING INVERSION ON NAV HOMOTOPY SEEDS. EQUAL PER-ARM BUDGET (80 ITERATIONS EACH), NO BANDIT. SCORES ARE THOSE OBSERVED AFTER THE FIRST 10-ITERATION BATCH: $s_\chi = -\ln \chi$; s_r IS THE CLIPPED FEASIBLE-IMPROVEMENT REWARD, WITH TIES AT THE CLIP CAP (50) BROKEN BY FIRST-SEEN ORDER AS IN THE SCHEDULER.

Seed	Oracle J^*	s_χ	s_r	Greedy- χ	Greedy- s_r
0	871.26	4.03	50.0		✓
1	538.91	6.33	50.0	✓	
2	538.88	5.83	6.35		

prebuilt binary rather than source, we validate its behaviour on a standard gate suite of five problems (pendulum swing-up, double-integrator reach, cartpole stabilisation, quadrotor hover, bicycle navigation) with three seeds each. All 15 instances converge to feasible solutions, and the resulting costs match the best open-source solver among Crocoddyl [3], Aligator, and ALTRO [2] to within 2% (Table III); the baseline scripts for the open-source solvers are part of the release. This establishes the host as a credible reference implementation; the paper's claims concern scheduling, not solver superiority.

C. Ranking-Flip Diagnostics

Lemma 1 establishes that, at the level of abstract cost sequences, local-improvement scores in S_{LI} can mis-rank seeds. We verify that this inversion occurs on real constrained-iLQR problems.

1) *Nav 2D equal-budget ranking flip*: On the Gate-aligned bicycle navigation benchmark with equal per-arm budgets (80 iterations each, no bandit), the first batch of 10 iterations already mis-ranks the seeds (Table IV): greedy selection by s_χ picks seed 1 and greedy selection by s_r picks seed 0, whereas the oracle ordering by terminal feasible cost prefers seed 2 ($J^* = 538.88$). Both scores therefore invert against the oracle.

2) *Contact-rich ranking flip (planar quadruped)*: The same early-vs-late inversion appears on the quadruped template (Table V). Cluster A (over-lifting gait) looks best after a short

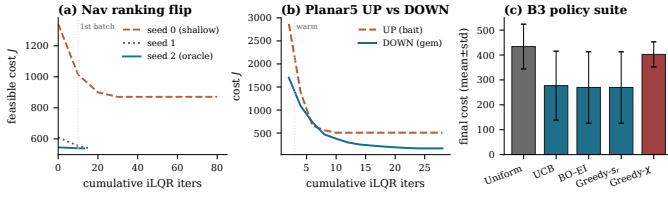


Fig. 2. Diagnostic and budgeting results. (a) Equal-budget Nav ranking flip: after the first batch, early J ranks disagree with the oracle terminal ordering (seed 2). (b) Planar5 equal-budget cost trajectories: UP improves fastest early but plateaus; DOWN overtakes and reaches a deeper basin. (c) Planar5 policy suite ($n = 5$): adaptive s_r policies beat Uniform; Greedy- χ remains worse despite full feasibility.

warm-up; cluster B (mild clearance) attains a lower terminal cost once more iterations are granted. The gate passes on all five RNG seeds.

These diagnostics instantiate the local-improvement incompleteness argument on three physically distinct systems (wheeled, legged, and articulated); they do not by themselves prove that any particular bandit policy dominates Uniform.

D. Seed-Budgeting Results

1) *B1: Nav2D-Hard*: Under the frozen protocol, adaptive allocation with s_r improves over Uniform on the dense-obstacle navigation benchmark (Table VI). Because many terminal iterates remain slightly infeasible at $\varepsilon = 10^{-3}$, we rank policies by feasible-first order (feasible $\succ$ lower violation $\succ$ lower cost). The adaptive bloc (UCB, BO-EI, Greedy- s_r) wins the feasible-first ranking against Uniform on all five RNG seeds; under the null hypothesis that the per-seed outcome is decided by chance, five wins in five seeds has probability 2^{-5} (one-sided sign test, $p \approx 0.03$). Greedy- χ attains a lower mean cost but at the cost of feasibility—it acts as a negative control that confirms the χ signal is not a reliable primary objective.

2) *B3: Planar5*: On the fully-feasible planar arm benchmark (Table VII), every run terminates without constraint violation, isolating score mis-ranking from feasibility entanglement. BO-EI and Greedy- s_r reach mean cost 269.7 ± 143.4 , approximately 38% lower than Uniform (433.9 ± 89.9); paired over the five RNG seeds, every adaptive policy attains lower cost than Uniform on every seed (one-sided sign test, $p = 2^{-5}$). Greedy- χ remains worse at 402.4 ± 50.5 despite full feasibility—a clean negative control.

Fig. 2 summarises the ranking-flip cost trajectories and the Planar5 policy comparison underlying Tables IV and VII.

3) *Computational Overhead*: GP inference in BO adds ~ 1.2 ms per scheduling round (Cholesky of an 8×8 matrix, $O(K^3)$), negligible compared to per-iteration iLQR cost (under 1 ms for B1 and B3). The overhead of χ computation adds no measurable latency. Per-run wall times are dominated by iLQR iterations: ~ 10 s for B1, ~ 0.3 s for B3.

The solver is implemented in C++17 with Eigen 3.4; experiments run on an AMD Ryzen 7 5800X (8-core, 3.8 GHz) with 32 GB RAM, Ubuntu 22.04.

IV. DISCUSSION AND LIMITATIONS

We have presented a feasibility-aware seed-budgeting framework for constrained iLQR, grounded in a ranking-

inversion analysis (Corollary 1) and a protocol that treats χ as a termination criterion rather than the primary reward. On two independent systems—Nav2D-Hard (wheeled navigation with dense obstacles) and Planar5 (5-DOF arm with UP/DOWN homotopies)—adaptive allocation with s_r improves over Uniform in feasible-first ranking. The negative control Greedy- χ either loses feasibility or attains higher cost, confirming that χ is not a reliable allocation signal. Ranking inversion is further demonstrated on a contact-rich quadruped template that is distinct from both the navigation and arm domains.

Several limitations warrant discussion. **Policy separation.** UCB, BO-EI, and Greedy- s_r remain tied on all three benchmarks; during warm-up the best seed is already identifiable by s_r , and the exploration bonus does not yield additional improvement on these libraries. Harder seed libraries with deeper local minima—where the greedy choice is actively misleading—would be needed to differentiate the policies. Similarly, the quadruped bandit suite is near-tied because warm-up already surfaces the better basin; the quadruped is cited as a ranking-flip diagnostic, not a policy-separation benchmark. On Nav2D-Hard, few runs return fully feasible incumbents at $\varepsilon = 10^{-3}$; we therefore report feasible-first comparisons rather than a single percentage gap. In one of the five seeds the adaptive winner is decided by lower violation alone despite higher cost; the feasible-first ordering is deliberately conservative about cost under infeasibility.

Scope of s_r . The reward $-\Delta J_{\text{feas}} - \lambda \text{viol}_+ + \eta\rho$ is a heuristic scalarisation. Its dominant term belongs to S_{LI} , so ranking inversion is guaranteed at the abstract level (Lemma 1). The violation penalty lacks a provable separation guarantee; its effectiveness depends on the constraint landscape and is verified empirically.

Generality. The framework supports arbitrary K , budget B , batch size b , and policy backends; the same scheduler can host any constrained iLQR solver without architectural changes. Extensions to DDP and SQP are straightforward.

V. CONCLUSION AND FUTURE WORK

We formulate multi-start constrained iLQR as *feasibility-aware seed budgeting*: allocate batches across warm-starts using diagnostics of feasible cost improvement, violation, and rate, with χ as a termination criterion rather than the primary reward. Three theoretical layers— χ non-monotonicity (Section II-C2), ranking inversion analysis (Corollary 1), and constructive inversion (Lemma 1)—establish that single-batch local-improvement scores are not universally optimal guides for seed allocation. Uniform, UCB, and GP-EI policies share an open, budget-exact scheduler hosted on a standard constrained iLQR stack.

On two independent systems (Nav2D-Hard and Planar5), adaptive allocation with s_r improves over Uniform under feasible-first ranking, while Greedy- χ either loses feasibility or attains higher cost. Ranking inversion is confirmed on three physically distinct domains—wheeled navigation, legged quadruped, and articulated arm—supporting the theoretical motivation across diverse robotics platforms. The ranking-flip evidence (Table IV) and the contact-rich diagnostic (Table V)

TABLE V

CONTACT-RICH RANKING-FLIP DIAGNOSTIC (PLANAR QUADRUPEL TEMPLATE, NO BANDIT). EACH OF $K = 4$ SEEDS RECEIVES WARM-UP 2 THEN +30 ILQR ITERATIONS. CLUSTER A (OVER-LIFT) IS BETTER AFTER WARM-UP; CLUSTER B (MILD CLEARANCE) ACHIEVES LOWER TERMINAL COST.

RNG	Best A @warm	Best A @late	Best B @late	Gate
42	22.2	22.2	7.98	PASS
43	22.9	22.9	9.39	PASS
44	26.0	26.0	7.95	PASS
45	23.1	23.1	7.98	PASS
46	22.1	22.1	13.1	PASS

TABLE VI

NAV2D-HARD SEED BUDGETING ($K = 8$, $B = 160$, $n = 5$). ADAPTIVE POLICIES (UCB, BO-EI, GREEDY- s_r) TIE WITH EACH OTHER AND WIN THE FEASIBLE-FIRST RANKING AGAINST UNIFORM ON 5/5 SEEDS.

Policy	Cost (mean $\pm$ std)	Feas. rate	Viol. (mean)
Uniform	2002.6 $\pm$ 435.6	0/5	0.306
UCB	1587.5 $\pm$ 852.4	2/5	0.083
BO-EI	1587.5 $\pm$ 852.4	2/5	0.083
Greedy- s_r	1587.5 $\pm$ 852.4	2/5	0.083
Greedy- χ	1418.7 $\pm$ 314.4	1/5	0.234

TABLE VII

PLANAR5 SEED BUDGETING ($K = 8$, $B = 80$, $n = 5$, ALL RUNS FEASIBLE). BO-EI AND GREEDY- s_r RECOVER THE DOWN HOMOTOPY; UNIFORM OFTEN STAYS ON THE UP BAIT; GREEDY- χ IS FEASIBLE BUT WORSE.

Policy	Cost (mean $\pm$ std)	Feas. rate
Uniform	433.9 $\pm$ 89.9	5/5
UCB	277.0 $\pm$ 138.4	5/5
BO-EI	269.7 $\pm$ 143.4	5/5
Greedy- s_r	269.7 $\pm$ 143.4	5/5
Greedy- χ	402.4 $\pm$ 50.5	5/5

instantiate the local-improvement incompleteness argument on real constrained-iLQR trajectories.

Future work includes richer seed generators, optional MOBO when speed and quality conflict, solver-agnostic hosting on alternative solvers via the same scheduler API, and on-line MPC receding-horizon re-budgeting. The benchmark definitions and all allocation traces are released at <https://github.com/jiesu-zju/optimal-solver>; the seed-budgeting scheduler and the constrained-iLQR host are provided as prebuilt binaries, and every reported experiment can be reproduced end-to-end with the exact command lines and seed parameters documented in the release.

REFERENCES

- [1] Y. Tassa, T. Erez, and E. Todorov, "Synthesis and stabilization of complex behaviors through online trajectory optimization," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2012, pp. 4906–4913.
- [2] T. A. Howell, B. E. Jackson, and Z. Manchester, "AL-TRO: A fast solver for constrained trajectory optimization," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2019, pp. 7674–7681.
- [3] C. Mastalli et al., "Crocodyl: An efficient and versatile framework for multi-contact optimal control," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2020, pp. 2536–2542.
- [4] Y. Zhu, J. J. Payne, S. Feng, and C. G. Atkeson, "Convergent iLQR for Safe Trajectory Planning and Control of Legged Robots," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.
- [5] M. D. McKay, R. J. Beckman, and W. J. Conover, "A comparison of three methods for selecting values of input variables in the analysis of output from a computer code," *Technometrics*, vol. 21, no. 2, pp. 239–245, 1979.
- [6] A. György and L. Kocsis, "Efficient multi-start strategies for local search algorithms," *J. Artif. Intell. Res.*, vol. 41, pp. 407–444, 2011.
- [7] S. Jain, D. Salau, and A. Agrawal, "A review of multi-start optimization algorithms," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 7, no. 5, pp. 2452–2457, 2019.
- [8] P. Auer, N. Cesa-Bianchi, and P. Fischer, "Finite-time analysis of the multiarmed bandit problem," *Mach. Learn.*, vol. 47, no. 2, pp. 235–256, 2002.
- [9] N. Srinivas, A. Krause, S. Kakade, and M. Seeger, "Gaussian process optimization in the bandit setting: No regret and experimental design," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 1015–1022.
- [10] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas, "Taking the human out of the loop: A review of Bayesian optimization," *Proc. IEEE*, vol. 104, no. 1, pp. 148–175, 2016.
- [11] W. Merkt, V. Ivan, and S. Vijayakumar, "Fast global motion planning via pre-computed motion primitives," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2021, pp. 7873–7879.
- [12] M. Posa, C. Cantū, and R. Tedrake, "A direct method for trajectory optimization of rigid bodies through contact," *Int. J. Robot. Res.*, vol. 33, no. 1, pp. 69–81, 2014.
- [13] J. Schulman et al., "Motion planning with sequential convex optimization and convex collision checking," *Int. J. Robot. Res.*, vol. 33, no. 9, pp. 1251–1270, 2014.
- [14] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. Cambridge, MA, USA: MIT Press, 2006.